

OPT-01: Single-Phase Transformer, ANSWER SHEET

A1 Q1 to Q20 (single-phase parts), answer-ready

not learning notes: exactly what to put on the paper, in order. Q11b and Q11c and Q15 (three-phase) live in OPT-02.

Q1 (verbatim): Explain the construction and working principle of a single-phase transformer with a neat diagram.

WRITE: construction (6 parts): (1) **core:** laminated CRGO silicon steel (0.35 to 0.5 mm insulated sheets), magnetic path for the flux; built core-type (windings around limbs) or shell-type (core around windings), with butt or lap interlocked joints. (2) **windings:** concentric on the core limbs, cylindrical in disc or helical or crossover form: LV winding inner (low insulation to core), HV outer; insulation class defines the conductor varnish and wrapping. (3) **insulation:** paper, mica, pressboard, enamel: class by temperature capability. (4) **tank** (sheet steel) with **transformer oil** for insulation and cooling, plus radiator tubes. (5) **conservator tank** (oil expansion cushion with air) and **silica-gel breather** (dry the entering air). (6) **bushings** (porcelain, HV out top and LV often at the bottom) and protection fittings (Buchholz relay on larger units). **WRITE:** working (3 lines): (1) AC supply to the primary gives an alternating mmf $N_1 i_1$ and an alternating mutual flux Φ_m in the core. (2) Faraday: this flux induces $e_1 = -N_1 \frac{d\Phi}{dt}$ (back emf) and $e_2 = -N_2 \frac{d\Phi}{dt}$ (load emf); the emf ratio is $\frac{E_1}{E_2} = \frac{N_1}{N_2}$. (3) With the secondary loaded, current I_2 flows and I_1 adjusts (mmf balance $N_1 I_1 \approx N_2 I_2$): power transfers by mutual induction, no electrical contact. **[DRAW]** core-type sectional view: laminated core, LV and HV concentric coils, tank, oil, conservator, breather, bushings. Label checklist: all 6 parts + Φ_m arrow + terminal marks.

Q2 (verbatim): Derive the EMF equation of a single-phase transformer and explain the significance of each parameter.

WRITE: derivation (the standard chain): (1) let the flux be $\Phi = \Phi_m \sin(\omega t)$ through a winding of N turns. (2) induced emf $e = -N \frac{d\Phi}{dt} = -N \omega \Phi_m \cos(\omega t)$. (3) peak value $E_{\max} = 2\pi f N \Phi_m$. (4) rms $E = \frac{E_{\max}}{\sqrt{2}} = \frac{2\pi}{\sqrt{2}} f N \Phi_m = 4.44 f N \Phi_m$. (5) generalise with the core area: $\Phi_m = B_m A$, giving $E = 4.44 f N B_m A$. (6) turns ratio line: $\frac{E_1}{E_2} = \frac{N_1}{N_2} = a$. **WRITE:** parameter significance (6 lines): f = supply frequency (Hz): induction needs changing flux. N = turns on that winding: emf scales with turns (the ratio knob). Φ_m = maximum mutual flux (Wb). B_m = maximum flux density in the core (T): set by the saturation knee of the steel. A = net iron cross-section (m^2 , net = stacking factor times gross). $4.44 = \frac{2\pi}{\sqrt{2}}$: the sinusoidal-waveform constant (4.0 for square waves, per the deck's variants note). E per turn = $\frac{E}{N} = 4.44 f \Phi_m$ is identical for both windings: the core design law.

Q3 (verbatim): Explain the ideal and practical transformer and discuss the assumptions made in an ideal transformer.

WRITE: ideal (the 5 assumptions): (1) no winding resistance (zero copper loss in coils). (2) no leakage flux: all flux links both windings (no leakage reactance drops). (3) lossless core: zero hysteresis and eddy (no iron loss). (4) infinite permeability: zero magnetising current needed to establish flux (mmf = 0). (5) perfect coupling (coefficient of coupling $k = 1$). **WRITE:** practical (the mapping, 5 lines): real copper resistance gives $I^2 R$ drop and heating; real leakage gives X_1 , X_2 drops and regulation; real steel gives

hysteresis and eddy loss (core loss, tested by OC); real finite permeability means a magnetising current of about 2 to 5 percent of full-load current flows at no load; coupling is essentially 1 in well-built cores (the textbook k -splits are pedagogy). Efficiency, regulation and the equivalent circuit (Q4) are exactly the corrections for these five imperfections. **DISTINGUISH:** ideal = five “zeros and infinities” (no R , no leakage, no iron loss, infinite μ , perfect coupling). practical = five real effects that generate the equivalent circuit’s six components.

Q4 (verbatim): Draw and explain the equivalent circuit of a single-phase transformer and identify all its components.

WRITE: component map (6 items): (1) R_1 : primary winding resistance (copper). (2) X_1 : primary leakage reactance (flux linking primary only). (3) R'_2 : secondary resistance referred to primary ($R_2 \times a^2$). (4) X'_2 : secondary leakage reactance referred ($X_2 \times a^2$). (5) R_0 : core-loss resistance across the supply (carries I_w , the core-loss current). (6) X_0 : magnetising reactance (carries I_m and creates Φ_m). The no-load current phasor splits: $\vec{I}_0 = \vec{I}_w + \vec{I}_m$ with $\cos \phi_0 = \frac{I_w}{I_0}$ small (0.1 to 0.25). **WRITE:** three forms (draw or state): exact circuit (shunt branch at the supply terminals), simplified (series $R_{eq} = R_1 + R'_2$ and $X_{eq} = X_1 + X'_2$ with the shunt branch at the input), approximate (shunt branch moved to the supply terminals and the drop computed on the load current alone: the workhorse of regulation and efficiency sums). Impedance transfer law: $Z' = Z \times a^2$. **[DRAW]** the standard approximate circuit: shunt R_0 , X_0 at the input, series R_{eq} , X_{eq} in the top lead, ideal 1: a transformer at the end with load.

Q5 (verbatim): Explain the methods used to determine the equivalent circuit parameters of a transformer.

WRITE: (1) The two standard tests: the **open-circuit test** finds the shunt branch (R_0 , X_0 , core loss); the **short-circuit test** finds the series branch (R_{eq} , X_{eq} , full-load copper loss). (2) Winding resistances can also be measured directly by direct-current resistance test (voltmeter-ammeter) at the machine terminals. (3) OC and SC together give every parameter without loading the transformer to its rating: that is their whole point. (4) State which side each result is referred to (the supply side used in the test), and use the a^2 law to move sides (Q10, Q9 give the formulas). (5) Verification line: the two tests’ losses add to the total loss used in efficiency sums.

Q6 (verbatim): Explain voltage regulation of a transformer and derive the expression for voltage regulation at different power factors.

WRITE: definition: regulation = the rise in secondary terminal voltage when the rated load (at stated power factor) is thrown off, the primary voltage held constant:

$$\text{regulation} = \frac{|V_{2,\text{no load}}| - |V_{2,\text{full load}}|}{|V_{2,\text{no load}}|} \quad (\text{deck convention})$$

[TRAP] three conventions exist in the literature (divide by $V_{2,\text{no load}}$, by $V_{2,\text{full load}}$, or using referred values). Name the convention you use and state the constant- V_1 condition: those two clauses carry a mark each. **WRITE:** derivation (phasor resolution, the chain to write): (1) refer all to the secondary (or primary): $V'_1 = V_2 + \vec{I}_2 (R_{eq} + jX_{eq})$ with $|V'_1|$ fixed at no-load value. (2) resolve along V_2 : for load p.f. $\cos \phi$ (lagging ϕ positive), the drop components along V_2 are $I_2 R_{eq} \cos \phi + I_2 X_{eq} \sin \phi$ (lagging) or $I_2 R_{eq} \cos \phi - I_2 X_{eq} \sin \phi$ (leading). (3) transverse components ($I_2 X \cos \phi \mp I_2 R \sin \phi$) are small second-order: their squares are negligible against V^2 (this neglect licenses Q8’s approximation). (4) exact squared form: $|V'_1|^2 = (V_2 + I_2 R \cos \phi \pm I_2 X \sin \phi)^2 + (I_2 X \cos \phi \mp I_2 R \sin \phi)^2$, then take the root and subtract. (5) the three-case summary:

$$\text{lagging: } \Delta V \approx I_2 (R_{eq} \cos \phi + X_{eq} \sin \phi) \quad \text{leading: } \Delta V \approx I_2 (R_{eq} \cos \phi - X_{eq} \sin \phi)$$

$$\text{unity: } \Delta V \approx I_2 R_{eq}$$

WRITE: the zero-regulation line: at a sufficiently leading power factor ΔV can hit zero when $\tan \phi = \frac{R_{eq}}{X_{eq}}$ (leading) and can even go negative (voltage rises on load): regulation range is negative to positive with leading p.f.

Q8 (verbatim): Derive the approximate expression for voltage regulation of a transformer for lagging and leading power factors.

WRITE: the derivation from Q6's step 4 (write it fully, it is its own question): (1) start at the exact squared equation above. (2) drop the transverse square (order of magnitude: drops are 5 to 10 percent of V , so their squares are 0.25 to 1 percent against V^2). (3) take the square root: $|V_1'| \approx V_2 + I_2 R \cos \phi \pm I_2 X \sin \phi$. (4) subtract and divide:

$$\text{regulation (lagging)} \approx \frac{I_2 (R_{eq} \cos \phi + X_{eq} \sin \phi)}{V_{2, \text{no load}}}$$

$$\text{regulation (leading)} \approx \frac{I_2 (R_{eq} \cos \phi - X_{eq} \sin \phi)}{V_{2, \text{no load}}}$$

WRITE: the three corollary forms (shortcut marks): (1) percent drop form: $\% \text{drop} \approx \%R \cos \phi \pm \%X \sin \phi$ where $\%R = \frac{I_2 R_{eq}}{V_2} \times 100$ and $\%X = \frac{I_2 X_{eq}}{V_2} \times 100$ at rated current. (2) percentage relations: $\%R = \frac{P_{cu, FL}}{S} \times 100$ (from the SC test watt reading!) and $\%Z = \frac{V_{sc}}{V_{rated}} \times 100$, $\%X = \sqrt{\%Z^2 - \%R^2}$: regulation from test data without circuit solving. (3) zero regulation: leading at $\tan \phi = \frac{R_{eq}}{X_{eq}}$ (Q6 step 5).

Q9 (verbatim): Explain the short-circuit test of a transformer with a neat circuit diagram and derive the equivalent impedance parameters.

WRITE: setup (5 points): (1) supply to the **high-voltage winding**, the **low-voltage winding short-circuited** by a thick link (the standard: because the LV current would be too large to measure, the HV current is small and safe; also less supply needed). (2) instruments on the HV side: voltmeter, ammeter, wattmeter (pressure coil before current coil). (3) the applied voltage is a small percentage of rated (2 to 12 percent) so the flux in the core is tiny and the **shunt branch draws negligible current**: the circuit is effectively just the series impedances. (4) raise the voltage slowly until the ammeter reads *rated HV current*: now the loss measured is the full-load copper loss. (5) this way full-load copper loss, equivalent impedance and regulation all follow from one small test. **WRITE:** derivation (the 3-parameter chain): with readings V_{sc} , I_{sc} , W_{sc} at the HV (primary) side:

$$Z_{eq} = \frac{V_{sc}}{I_{sc}}, \quad R_{eq} = \frac{W_{sc}}{I_{sc}^2}, \quad X_{eq} = \sqrt{Z_{eq}^2 - R_{eq}^2}$$

and the load-loss law: P_{cu} at any current $= I^2 R_{eq}$, so at load fraction x : $P_{cu} = x^2 P_{cu, FL}$. To move to the secondary: multiply the referred values by $\frac{1}{a^2}$. **WRITE:** precautions (4): (1) use a thick shorting link: thin links add unmeasured resistance and inflate R_{eq} . (2) measure on the HV side and record which side the numbers are referred to. (3) wait for temperature stabilisation if repeated (resistance drifts). (4) the wattmeter reading is small and $\cos \phi$ is tiny (impedance angle near 90°): use a low-power-factor wattmeter.

Q10 (verbatim): Explain the open-circuit test of a transformer and show how core loss and no-load parameters are determined.

WRITE: setup (5 points): (1) supply to the **low-voltage winding** at rated voltage and frequency, the **HV winding open** (the standard: rated voltage is small and safe on the LV side and the small no-load current is easily measured on the LV; on the HV side the ammeter would read too little). (2) three instruments

on the LV side: voltmeter, ammeter (low-reading), wattmeter (low-power-factor type). (3) at no load the secondary current is zero so the series drops vanish: the wattmeter reads exactly the **core (iron) loss** at rated voltage (hysteresis + eddy, scale-free from load). (4) the applied voltage fixes Φ_m and B_m (from the EMF equation), so the reading is the rated-flux iron loss. (5) copper loss at no load ($I_0^2 R_1$) is tiny and neglected (or estimated). **WRITE:** derivation (the parameter chain): readings V_0 , I_0 , W_0 at the LV side:

$$\cos \phi_0 = \frac{W_0}{V_0 I_0} \quad I_w = I_0 \cos \phi_0 \text{ (core-loss component)}, \quad I_m = I_0 \sin \phi_0 \text{ (magnetising component)}$$

$$R_0 = \frac{V_0}{I_w} = \frac{V_0^2}{W_0} \text{ (two routes: use both and cross-check)} \quad X_0 = \frac{V_0}{I_m}$$

WRITE: outputs to state: core loss $P_i = W_0$ (feeds the efficiency sums), and the no-load branch R_0 , X_0 on the LV side (multiply by a^2 for the HV side). The no-load power factor $\cos \phi_0$ is small (0.1 to 0.25): that is why I_m dominates I_0 and the magnetising role question (Q17) matters.

Q7 (verbatim): Derive the expression for transformer efficiency and explain the condition for maximum efficiency.

WRITE: derivation (the chain): (1) $\eta = \frac{\text{output}}{\text{output} + \text{losses}} = \frac{V_2 I_2 \cos \phi}{V_2 I_2 \cos \phi + P_i + P_{cu}}$. (2) at load fraction x of rated kVA S : $P_{cu} = x^2 P_{cu,FL}$ (the scaling law) and P_i stays constant at rated V , f :

$$\eta(x) = \frac{x S \cos \phi}{x S \cos \phi + P_i + x^2 P_{cu,FL}}$$

(3) maximum: differentiate wrt I_2 (or x). With $k = V_2 \cos \phi$: $\eta = \frac{k I_2}{k I_2 + P_i + I_2^2 R_{eq2}}$; the quotient rule gives $\frac{d\eta}{dI_2} \propto P_i - I_2^2 R_{eq2}$. (4) set to zero: $P_i = I_2^2 R_{eq2} = P_{cu}$: **maximum efficiency when copper loss equals iron loss**. (5) the load point and the peak value:

$$I_{2,\max\eta} = \sqrt{\frac{P_i}{R_{eq2}}} \quad x_{\max\eta} = \sqrt{\frac{P_i}{P_{cu,FL}}} \quad \eta_{\max} = \frac{x S \cos \phi}{x S \cos \phi + 2P_i}$$

(6) why it lands below rated load: design P_i slightly under $P_{cu,FL}$ so the peak sits where the transformer actually runs (distribution units: 50 to 70 percent loading; power transformers: nearer 80 to 90). (7) $\cos \phi$ scales the output and the peak's height but not the *load fraction* x of the peak.

Q11a (verbatim, part 1 of the 3-parter): Explain the condition for maximum efficiency of a transformer and derive the corresponding expression.

WRITE: short self-contained version (destination identical to Q7): (1) losses: constant iron P_i + load-squared copper $x^2 P_{cu,FL}$. (2) write $\eta(x)$ as in Q7 and set $\frac{d\eta}{dx} = 0$: the algebra lands on $P_i = x^2 P_{cu,FL}$. (3) state the condition: **copper loss equals iron loss** at the load of maximum efficiency. (4) then $x = \sqrt{\frac{P_i}{P_{cu,FL}}}$ and

$$\eta_{\max} = \frac{x S \cos \phi}{x S \cos \phi + 2P_i} = \frac{S \cos \phi}{S \cos \phi + 2\sqrt{P_i P_{cu,FL}}}$$

(5) worked shape: $P_i = 350 \text{ W}$, $P_{cu,FL} = 400 \text{ W} \Rightarrow x = 0.935$ (93.5 percent load) and $\eta_{\max} = 97.09$ percent at unity p.f. (96.40 percent at 0.8: the peak *load* does not move with p.f.).

Q12 (verbatim): What are the necessary conditions for the parallel operation of transformers? Explain each condition.

WRITE: the split answer (essential first, then desirable, explaining each): **essential (without these the bank is destroyed or dead):** (1) **same polarity:** opposite polarity closes a dead short across the pair at the instant of connection: massive circulating current and burnout. (2) **same voltage ratio (and phase sequence in 3-phase):** unequal ratios leave a voltage difference around the closed loop driving a circulating current even at no load (Q13). (3) **same frequency** (inherent, one line). **desirable (without these it runs but shares badly):** (4) **same percent (per-unit) impedance magnitudes** on each transformer's own rating: currents then divide inversely to impedance exactly in proportion to rating: neither unit overloads before the bank is full. (5) **same impedance angles (same $\frac{X}{R}$ ratio):** otherwise the two share currents of different phase angles: one carries more kVA and worse power factor than its share. (6) **identical construction and size** where possible: keeps resistances, reactances and temperature rise matched over the load range.

Q13 (verbatim): Explain the causes and effects of circulating current in transformers operating in parallel and suggest methods to avoid it.

WRITE: causes (3 points): (1) unequal voltage ratios: the two induced secondary emfs do not match, and around the closed parallel loop the unbalanced difference drives a current with no load attached at all:

$$I_{\text{circ}} = \frac{\vec{E}_A - \vec{E}_B}{\vec{Z}_A + \vec{Z}_B}$$

(2) wrong polarity (extreme case: the emfs add around the loop instead of opposing: short-circuit current). (3) unequal per-unit impedances aggravate the sharing problem once any emf mismatch exists (the circulating current adds to one unit and subtracts from the other). **WRITE:** effects (5 points): (1) copper loss at *no load*: wasted energy and heat with zero useful output. (2) reduced kVA capacity: part of each winding's rating is consumed by the circulating current before any load is served. (3) distorted load sharing: one unit overloads while the other under-loads at a much lower bank limit. (4) temperature rise and insulation aging above design. (5) protection may trip (overcurrent relays see the sum). **WRITE:** remedies (4 points): (1) manufacture to tight turns-ratio tolerances. (2) correct residual ratio error with **tap changers** (adjust taps until the circulating current vanishes). (3) keep equal per-unit impedances so any small emf mismatch meets enough impedance to limit the circulating current. (4) verify polarity and connection marks (H_1 with H_1 , X_1 with X_1) before closing the bank.

Q14 (verbatim): Explain the load sharing between transformers connected in parallel and derive the condition for proper load sharing.

WRITE: derivation (4 steps): (1) two units in parallel across one bus: equal voltage across both branches:

$$\vec{I}_A \vec{Z}_A = \vec{I}_B \vec{Z}_B, \text{ and KCL: } \vec{I}_L = \vec{I}_A + \vec{I}_B. \quad (2) \text{ solve (phasor divider): } \vec{I}_A = \vec{I}_L \frac{\vec{Z}_B}{\vec{Z}_A + \vec{Z}_B} \text{ and } \vec{I}_B =$$

$$\vec{I}_L \frac{\vec{Z}_A}{\vec{Z}_A + \vec{Z}_B}. \quad (3) \text{ magnitudes, same impedance angles: } \frac{I_A}{I_B} = \frac{Z_B}{Z_A} \text{ (inverse impedance sharing) and the}$$

kVA shares follow the currents (same voltage): $\frac{S_A}{S_B} = \frac{Z_B}{Z_A}$. (4) with unequal ratios add the circulating

$$\text{term to each (the Q13 current, opposite signs): } \vec{I}_A = \vec{I}_L \frac{\vec{Z}_B}{\vec{Z}_A + \vec{Z}_B} + \vec{I}_{\text{circ}}, \quad \vec{I}_B = \vec{I}_L \frac{\vec{Z}_A}{\vec{Z}_A + \vec{Z}_B} - \vec{I}_{\text{circ}}.$$

WRITE: proper (proportional) sharing condition: each unit's kVA must reach its nameplate together, i.e.

$$\frac{S_A}{S_B} = \frac{S_{A,\text{rated}}}{S_{B,\text{rated}}}, \text{ which by step 3 requires } \boxed{\frac{Z_A}{Z_B} = \frac{S_{B,\text{rated}}}{S_{A,\text{rated}}}} \text{ with equal angles: in per-unit terms: equal}$$

percent impedance on own base. Bigger unit must own the smaller impedance. **WRITE:** the worked

shape (D1-07): $Z_A = 0.5 + j3 \Omega$, $Z_B = 0.6 + j10 \Omega$, 100 kW at 0.8 p.f. (125 kVA): shares 0.768 and 0.233 of the current phasors (angles 1.4° and -4.6°): 96 kVA and 29 kVA: a 77 : 23 split showing the unequal-impedance distortion live.

Q16 (verbatim): Why does a transformer not work on DC supply? Explain using the principle of electromagnetic induction.

WRITE: (1) induction law: $e = -N \frac{d\Phi}{dt}$: only *changing* flux induces emf. (2) direct current at steady state produces a **constant** flux: $\frac{d\Phi}{dt} = 0$, so $E_2 = 0$: no output whatever the current in the primary. (3) with no back emf ($E_1 = 0$ too), the primary becomes a pure resistance: $I_1 = \frac{V}{R_1}$ with R_1 tiny: the current is enormous, the winding overheats and burns (this is the DC short-circuit case). (4) also no reactive power transfer mechanism: the whole transformer action (mmf balance with induced emfs) rests on alternating flux. (5) the transient caveat line: at the *instant* of switching DC on there is a brief induction spike ($\frac{d\Phi}{dt}$ exists once), after which flux is constant and the collapse to the resistive state follows.

Q17 (verbatim): Why does a transformer draw a small current even when its secondary is open-circuited? Explain the role of the magnetizing component.

WRITE: (1) with the secondary open, no load power is drawn, but the primary still must **create the alternating mutual flux** Φ_m that makes the back emf $E_1 \approx V_1$: an mmf is needed even at no load because the steel has finite permeability. (2) that current is the **no-load current** I_0 : only about 2 to 5 percent of full-load current in a normal transformer. (3) it splits into two components: $\vec{I}_0 = \vec{I}_w + \vec{I}_m$: the **magnetising component** I_m (wattless, in quadrature with the flux) builds and sustains the flux Φ_m ; the **core-loss (working) component** I_w (in phase with the supply voltage) feeds the hysteresis and eddy losses in the core. (4) the magnetising role sentence: I_m is the current whose ampere-turns $N_1 I_m$ balance the core's reluctance drop and establish B_m (the mmf across the magnetic circuit is $H\ell$ and the steel needs real current for it). (5) since $\cos \phi_0$ is small (0.1 to 0.25), I_m dominates I_0 : the no-load current is mostly magnetising, which is exactly why X_0 (not R_0) is the big-impedance branch at no load.

Q18 (verbatim): Why is the transformer core laminated?

WRITE: (1) alternating flux induces emfs *in the core iron itself* (it is a conductor in a changing field): these drive circulating currents in closed loops in the core cross-section: **eddy currents**. (2) eddy currents dissipate $I^2 R$ heat inside the core: wasted power and dangerous hot spots (and they distort flux). (3) eddy loss magnitude: $P_e = k_e f^2 B_m^2 t^2$ (per unit mass): it scales with the *square* of the conductor (sheet) thickness t . (4) laminating: build the core of thin insulated sheets (0.35 to 0.5 mm, varnish or oxide coated) stacked to the required area: each eddy loop is confined to one sheet, its path cross-section shrinks, its resistance rises, and the loop area collapses: the loss falls as t^2 . (5) halving the sheet thickness quarters the eddy loss: the whole lamination payoff in one sentence. (6) trade-offs to state: insulation varnish costs stacking factor (net iron area about 0.9 to 0.95 of gross: slightly larger core for the same Φ) and adds manufacturing cost. (7) companion line: the other iron loss, hysteresis, is cut by *material* choice (CRGO silicon steel, narrow loop) not by lamination: keep the two losses and their remedies distinct.

Q19 (verbatim): Why are transformers rated in kVA rather than kW? Justify your answer considering transformer losses.

WRITE: (1) rating = a **thermal promise**: the output is limited by heat, which is limited by losses. (2) the two loss families track two different quantities: core (iron) loss is fixed by *voltage* (V sets Φ_m and B_m : $P_h = k_h f B_m^{1.6}$, $P_e = k_e f^2 B_m^2 t^2$); copper loss is fixed by *current* ($P_{cu} = I^2 R$). (3) neither loss knows the

load power factor: at the same rated V and I the heating is identical whether the load is unity or 0.5 p.f. (4) $\text{kW} = \text{kVA} \times \cos \phi$ would vary with the *user's* power factor at identical heating: a kW rating would be wrong at one p.f. or the other. (5) the only p.f.-independent quantity that captures both limits is the apparent power $S = VI$: hence **rated kVA** $= \frac{V_{\text{rated}} I_{\text{rated}}}{1000}$. (6) illustration: one unit happily serving 100 kW at unity p.f. (100 kVA) would be fully loaded at 50 kW when $\cos \phi = 0.5$ (still 100 kVA, same I , same heating). (7) efficiency and regulation *do* depend on p.f. (they are always asked “at 0.8 lag”), but the *rating* is thermal: kVA.

Q20 (verbatim): Explain the losses in a transformer and discuss methods for reducing each type of loss.

WRITE: loss map (4 items) with one reduction bundle each: (1) **copper loss** (I^2R in both windings, $P_{\text{cu}} = x^2 P_{\text{cu,FL}}$ with load): reduce by thicker (lower-resistance) conductors, lower-resistivity copper, shorter mean turn length. (2) **hysteresis loss** ($P_h = k_h f B_m^{1.6}$: energy spent re-orienting domains each cycle): reduce by narrow-loop material (CRGO silicon steel), operating B_m below the saturation knee, lower frequency if variable. (3) **eddy loss** ($P_e = k_e f^2 B_m^2 t^2$: circulating currents in the iron): reduce by lamination (thin t , the t^2 lever), insulated sheets, higher-resistivity steel alloy. (4) **stray loss** (leakage-flux eddy currents in windings, tank, clamps): reduce by interleaved winding, non-magnetic tank parts, magnetic shielding, proper conductor transposition. **WRITE:** the closing laws (3 lines): at constant supply, iron loss is **constant** and copper loss varies with the **square** of load current. Total loss curve $= P_i + x^2 P_{\text{cu,FL}}$: the drop-versus-loss trap (V -drop scales with x , copper loss with x^2). Static machine: no mechanical (friction and windage) loss at all: that line distinguishes this answer from the DC machine version. Efficiency peak: variable = constant losses (Q7).

NUMERIC ANSWER STRIP (all 17 verified against report 05)

WE-01 100 kVA SC: $Z_{\text{eq}} = 5.8 \Omega$, $R_{\text{eq}} = 1.2 \Omega$, $X_{\text{eq}} = 5.67 \Omega$; η : 98.21 (FL u.p.), 98.03 (FL 0.8), 98.62 (half u.p.). **WE-02** 20 kVA, 2000 V to 200 V: $R_{\text{eq2}} = 0.909 \Omega$; η : 93.74 (FL 0.8), 92.75 (half), max 93.82 at 93.5 percent load. **WE-03** [dangling: continuation missing, unanswerable]. **WE-04** 100 kVA, 11000 V to 220 V: emf per turn 25 V; $N_1 = 440$, $N_2 = 50$. **WE-05** 200 V 50 Hz 100 turns: $\Phi_m = 0.08$ Wb. **WE-06** $E_1 = 5000$, $N_1 = 500$, $N_2 = 50$: $E_2 = 500$ V [errata E5 and E6: source prints 1200 V and a phantom 0.026 T: wrong]. **WE-07** 50 kVA, 2200 V to 250 V: HV-referred $R_{\text{eq}} = 27.23$, $X_{\text{eq}} = 27.06$ (Z 38.3 class), LV-referred $R_{\text{eq2}} = 0.351$, $X_{\text{eq2}} = 0.349$ [errata E4: label which side every number lives on]. **D1-01** 100 kVA, 2000 V to 400 V (R 0.04, X 0.07 p.u. class): reg 3.42 percent (0.8 lag), 2.53 (0.8 lead), 1.39 (unity); zero regulation at 0.981 leading. **D1-02** 50 kVA, 2000 V to 500 V: loaded secondary $V_2 = 377.7$ V. **D1-05** 5 kVA, 2000 V to 100 V OC 100 V 25 W 0.7 A, SC 60 V 180 W 2.5 A: route A $Z_{01} = 391.7$, $R_{01} = 80$, $X_{01} = 383.4$; route B 392, 88, 382 [errata E14: both routes defensible: state the shorting assumption]; η 96.68 (FL 0.8), 97.45 (FL u.p.), 97.62 (max); reg 6.17 percent (0.8 lag). **D1-06** 25 kVA, 2200 V to 250 V OC 250 150 W 3 A, SC 90 550 W 10 A: $Z_{02} = 4.686 \Omega$, $R_{02} = 1.35 \Omega$, $X_{02} = 4.485 \Omega$ (values on LV side: E4 rule); reg 4.82 percent (0.8 lag) and 3.33 percent (0.8 lead); η 95.58 percent (0.8 lag). **WE-08** 40 kVA, 6600 V to 250 V: drop 9.22 V at 0.8 lag = 3.69 percent (deck convention) and 3.56 percent (LV full-load convention) [denominator war: name yours]. **WE-09** 63 kVA, 2200 V to 400 V: drop 8.25 V = 3.75 percent fall at 0.8 lag; at 0.8 *lead*: 2.2 percent *rise* [errata E9: the source's labels confuse percent with volts]. **D1-07** see Q14 worked shape.

FORMULA STRIP

$$E = 4.44fN\Phi_m = 4.44fNB_mA. \quad \frac{E_1}{E_2} = \frac{N_1}{N_2} = a, \quad Z' = Za^2. \quad Z_{\text{eq}} = \frac{V_{\text{sc}}}{I_{\text{sc}}}, \quad R_{\text{eq}} = \frac{W_{\text{sc}}}{I_{\text{sc}}^2}, \quad X_{\text{eq}} = \sqrt{Z_{\text{eq}}^2 - R_{\text{eq}}^2}. \quad \cos \phi_0 = \frac{W_0}{V_0 I_0}, \quad I_w = I_0 \cos \phi_0, \quad I_m = I_0 \sin \phi_0, \quad R_0 = \frac{V_0^2}{W_0}, \quad X_0 = \frac{V_0}{I_m}. \quad \text{Reg} \approx \frac{I_2(R_{\text{eq}} \cos \phi \pm X_{\text{eq}} \sin \phi)}{V_{2,\text{nl}}};$$

$$\%drop = \%R \cos \phi \pm \%X \sin \phi. \quad \eta = \frac{xS \cos \phi}{xS \cos \phi + P_i + x^2 P_{cu,FL}}; \text{ max at } P_i = P_{cu}, x = \sqrt{\frac{P_i}{P_{cu,FL}}}. \text{ Parallel:}$$

$$\frac{I_A}{I_B} = \frac{Z_B}{Z_A}; \text{ proper sharing } \frac{Z_A}{Z_B} = \frac{S_{B,rated}}{S_{A,rated}}. \quad I_{circ} = \frac{\vec{E}_A - \vec{E}_B}{\vec{Z}_A + \vec{Z}_B}. \quad P_h = k_h f B_m^{1.6}, P_e = k_e f^2 B_m^2 t^2.$$

MEMORY DUMP (write cold)

$E = 4.44 f N B_m A$; per turn common. Ideal = five zeros and infinities μ . SC on HV: series params. OC on LV: R_0 , X_0 , core loss, $\cos \phi_0$ split. Regulation: constant V_1 , name the denominator; lag ($R \cos \phi + X \sin \phi$), lead minus; zero reg at $\tan \phi = \frac{R}{X}$. Efficiency: P_{cu} scales x^2 , P_i flat; max at equal losses, $x = \sqrt{\frac{P_i}{P_{cu,FL}}}$. Parallel: same polarity and ratio (essential), equal per-unit R , X , angle (sharing). Circulating current: ΔE over $Z_A + Z_B$, kills at no load. No DC: constant flux, no E , primary shorts. No-load current: I_w (loss) + I_m (flux), 2 to 5 percent. Laminated: eddy dies as t^2 . kVA: losses track V and I , not $\cos \phi$. Losses: copper, hysteresis, eddy, stray; static machine: no mechanical loss.